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Semiconductor fabrication apparatus

Abstract

A semiconductor fabrication apparatus comprising a light source configured to emit light, a substrate stage arranged to receive a substrate exposed to the emitted light, a reticle arranged between the substrate stage and the light source, and a reticle stage arranged to receive the reticle. The reticle stage including a lower plate, an upper plate arranged above the lower plate, an actuator connected to the lower plate configured to move in a direction parallel to the upper plate, a first cable slab arranged between the upper plate and the lower plate and connected to one side of the actuator, and a first cable cover that surrounds an outer periphery of the first cable slab and contacts the lower plate when the first cable slab becomes bent.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2022-0015922 filed on Feb. 8, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) Embodiments of the present inventive concept relate to a semiconductor fabrication apparatus, and more particularly, to a semiconductor fabrication apparatus used for photolithography.

DISCUSSION OF RELATED ART

(3) In general, a photolithography process may include a coating process, an exposure process, and a developing process. The exposure process may be a process that transports an image pattern onto a photoresist on a substrate. With an increase in the integration of semiconductor devices, research on the exposure process is being conducted to reduce a wavelength of light used in the exposure process to transport an image pattern onto a photoresist on a substrate. A typical semiconductor fabrication apparatus may include a reticle stage and a substrate stage. Each of the reticle stage and the substrate stage may have a cable slab connected to a controller.

SUMMARY

(4) Embodiments of the present inventive concept provide a semiconductor fabrication apparatus capable of preventing or minimizing abrasion of cable slab.

(5) According to some embodiments of the present inventive concept, a substrate fabrication apparatus may comprise: a light source configured to emit light; a substrate stage that receives a substrate exposed to the light; a reticle arranged between the substrate stage and the light source; and a reticle stage that receives the reticle. The reticle stage may include: a lower plate; an upper plate above the lower plate; an actuator connected to the lower plate, arranged to move in a direction parallel to the upper plate; a first cable slab arranged between the upper plate and the lower plate, the first cable slab being connected to one side of the actuator; and a first cable cover that surrounds an outer periphery of the first cable slab and contacts the lower plate when the first cable slab is bent.

(6) According to some embodiments of the present inventive concept, a semiconductor fabrication apparatus may comprise: a chamber; a light source in the chamber configured to emit light; a substrate stage arranged to receive a substrate exposed to the light; a reticle arranged between the substrate stage and the light source; and a reticle stage that receives the reticle. The substrate stage may include: a lower plate; an upper plate on the lower plate; an actuator connected to the lower plate, actuator moving in a direction parallel to the upper plate; a cable slab arranged between the upper plate and the lower plate, the cable slab being connected to one side of the actuator; and a cable cover that surrounds an outer periphery of the cable slab, the cable cover contacting the lower plate when the cable slab is bent.

(7) According to some embodiments of the present inventive concept, a semiconductor fabrication apparatus may comprise: a light source configured to emit light; a substrate stage that receives a substrate exposed to the light; a reticle between the substrate stage and the light source; and a reticle stage that receives the reticle. The reticle stage may include: a first lower plate; a first upper plate on the first lower plate; a first actuator connected to the first lower plate and moving in a direction parallel to the first upper plate; a first cable slab between the first upper plate and the first lower plate and connected to one side of the first actuator; a first cable cover that surrounds an outer periphery of the first cable slab and contacts the first lower plate when the first cable slab is bent; a second cable slab between the first upper plate and the first lower plate and connected to another side of the first actuator; a second cable cover that surrounds an outer periphery of the second cable slab and contacts the first lower plate when the second cable slab is bent; and a roller vehicle on the first cable slab and the second cable slab, the roller vehicle moving together with the first actuator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other features of the inventive concept will be more clearly understood from the

following detailed description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates a cross-sectional view of a semiconductor fabrication apparatus according to an embodiment of the present inventive concept.

(3) FIGS. 2 and 3 illustrate a plan view and a cross-sectional view of a reticle stage of the semiconductor fabrication apparatus as depicted in FIG. 1.

(4) FIGS. 4 and 5 respectively illustrate a development view and a plan view of a first cable cover of the reticle stage as depicted in FIGS. 2 and 3.

(5) FIG. 6 illustrates a cross-sectional view of a reticle stage of the semiconductor fabrication apparatus as depicted in FIG. 1.

(6) FIG. 7 illustrates a cross-sectional view of a roller vehicle of the reticle stage as depicted in FIG. 6.

(7) FIG. 8 illustrates a plan view of a reticle stage of the semiconductor fabrication apparatus as depicted in FIG. 1.

(8) FIG. 9 illustrates a cross-sectional view of a semiconductor fabrication apparatus according to an embodiment of the present inventive concept.

(9) FIG. 10 illustrates a cross-sectional view of a substrate stage of the semiconductor fabrication apparatus as depicted in FIG. 9.

DETAIL DESCRIPTION OF EMBODIMENTS

(10) Exemplary embodiments of the present inventive concept will be described more fully hereinafter with reference to the accompanying drawings. Like reference numerals may refer to like elements throughout the accompanying drawings.

(11) FIG. 1 shows an example of a semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concept.

(12) Referring to FIG. 1, a semiconductor fabrication apparatus **100** of the present inventive concept may be embodied, for example, as a scanner exposure device. Alternatively, the semiconductor fabrication apparatus **100** may be embodied as a stepper exposure device. For example, the semiconductor fabrication apparatus **100** may be an ArF exposure device or a KrF exposure device. According to an embodiment, the semiconductor fabrication apparatus **100** may include a light source **10**, a substrate stage **20**, an illumination optical system **30**, a projection optical system **40**, a reticle **50**, and a reticle stage **60**.

(13) The light source **10** may produce and provide light **110** to be emitted towards a substrate W. In an embodiment, the light **110** may include excimer laser light or deep ultraviolet light. For example, the light **110** may have a wavelength of 193 nm (ArF), a wavelength of 248 nm (KrF), a wavelength of 365 nm (I-line), or a wavelength of 436 nm (G-line).

(14) The substrate stage **20** may be provided below the light source **10** and the projection optical system **40**. The substrate stage **20** may receive a substrate W. The substrate stage **20** may move the substrate W in a horizontal direction. The substrate W may include a silicon wafer, but the present inventive concept is not limited thereto.

(15) The illumination optical system **30** may be provided between the light source **10** and the substrate stage **20**. The illumination optical system **30** may be provided between the light source **10** and the reticle **50**. For example, the illumination optical system **30** may include an aperture and a lens. The illumination optical system **30** may be configured to carry out an enlargement projection of the light **110** and/or a reduction projection of the light **110**. Alternatively, the illumination optical system **30** may include a mirror, but the present inventive concept is not limited thereto. The illumination optical system **30** may change a path of the light **110** and collimate the light **110**.

(16) The projection optical system **40** may be provided between the reticle **50** and the substrate stage **20**. The projection optical system **40** may focus light **110** on the substrate W, whereby the focused light **110** penetrates the reticle **50**. The projection optical system **40** may include an objective lens (e.g., convex lens).

(17) Reticle **50** may be provided between the illumination optical system **30** and the projection

optical system **40**. The reticle **50** may be provided in the reticle stage **60**. For example, the reticle **50** may be a transmissive photomask. The reticle **50** may project the light **110** on the substrate **W**.

(18) The reticle stage **60** may receive the reticle **50**. The reticle stage **60** may allow the reticle **50** to move in response to a control signal of a controller (See **102** of FIG. **3**).

(19) FIGS. **2** and **3** illustrate the reticle stage **60** as depicted in FIG. **1**, according to one embodiment.

(20) Referring to FIGS. **2** and **3**, the reticle stage **60** may include first lower plates **62**, first upper plates **64**, a first actuator **66**, a first cable slab **72**, a second cable slab **74**, a first cable cover **82**, and a second cable cover **84**.

(21) The first lower plates **62** may be correspondingly provided below the first upper plates **64**. Although not shown in FIGS. **2** and **3**, the first lower plates **62** may be provided on opposite sides of the first actuator **66**. The first lower plate **62** may guide a lower portion of the second cable cover **84**, the first cable cover **82**, the second cable slab **74**, and the first cable slab **72**. The first lower plate **62** may be a lower guide plate. The first lower plate **62** may move along the first actuator **66**. Alternatively, the first lower plate **62** may stop independently of the first actuator **66**, but the present inventive concept is not limited thereto.

(22) The first upper plates **64** may be provided on opposite sides of the first actuator **66**. The first upper plates **64** may be correspondingly provided on the first lower plates **62**. The first upper plate **64** may be spaced apart from the first lower plate **62**. The first upper plate **64** may guide an upper portion of the second cable cover **84**, the first cable cover **82**, the second cable slab **74**, and the first cable slab **72**. The first upper plate **64** may be an upper guide plate.

(23) The first actuator **66** may be connected to the first lower plate **62**. The first actuator **66** may move along a guide rail (not shown). The guide rail may be parallel to the first lower plate **62** and the first upper plate **64**. For example, the first actuator **66** may include a first long stroke actuator **65** and a first short stroke actuator **67**.

(24) The first long stroke actuator **65** may be provided below the first short stroke actuator **67**. As shown in the plan view of FIG. **2**, the first long stroke actuator **65** may be wider than the first short stroke actuator **67**. The first long stroke actuator **65** may drive the first short stroke actuator **67** to move in a horizontal direction. The first short stroke actuator **67** may be provided on a center of the first long stroke actuator **65**. The first short stroke actuator **67** may have a reticle chuck **69** in a center thereof.

(25) The reticle **50** may be provided in the reticle chuck **69**. The first short stroke actuator **67** may drive the reticle **50** to move in a horizontal direction (e.g., X and Y directions).

(26) The first cable slab **72**, as shown in FIG. **3**, may be provided between the first lower plate **62** and the first upper plate **64**. A controller **102** may be connected through the first cable slab **72** to one side of the first long stroke actuator **65** of the first actuator **66**. The first cable slab **72** may be curved or bent.

(27) Although not shown, each of the first and second cable slabs **72** and **74** may include conductive lines and a cover that surrounds outer peripheries of the conductive lines.

(28) The second cable slab **74** may be provided between the first lower plate **62** and the first upper plate **64**. A controller **102** may be connected through the second cable slab **74** to another side of the first long stroke actuator **65** of the first actuator **66**. The second cable slab **74** may be bent in a direction toward the first cable slab **72**.

(29) The first cable cover **82** may be provided on and below the first cable slab **72**. The first cable cover **82** may protect the first cable slab **72**. When the first cable slab **72** bends, the first cable cover **82** may prevent or minimize abrasion of the first cable slab **72** against the first lower plate **62**. The first cable cover **82** may include Teflon. The first cable cover **82** may not be attached to, but instead be separated from the first cable slab **72**. For example, the first cable cover **82** may have a width of about 280 mm and a length of about 680 mm.

(30) Each of the first and second cable covers **82**, **84** may include a cable jacket and a cable carrier,

but the present inventive concept is not limited thereto. In addition, each of the first and second cable covers **82**, **84** may be provided on upper and lower sides of a tube slab (not shown), thereby protecting the tube slab.

(31) The second cable cover **84** may be provided on and below the second cable slab **74**. The second cable cover **84** may protect the second cable slab **74**. When the second cable slab **74** bends, the second cable cover **84** may prevent or minimize abrasion of the second cable slab **74** against the first lower plate **62**. The second cable cover **84** may be the same as the first cable cover **82**. For example, the second cable cover **84** may include Teflon.

(32) Each of the first and second cable slabs **72** and **74** in the first and second cable covers **82** and **84** may have a stack structure or a parallel structure. Likewise, the tube slab may have a stack or parallel structure in the first cable cover **82** and the second cable cover **84**.

(33) Although not shown in FIGS. **2** and **3**, in one embodiment, the first cable cover **82** and the second cable cover **84** may not be coupled to fixing clamps of the first cable slab **72** and the second cable slab.

(34) FIGS. **4** and **5** illustrate development and plan views of the first cable cover **82** depicted in FIGS. **2** and **3**.

(35) Referring to FIGS. **4** and **5**, the first cable cover **82** may include a lower cover **81** and an upper cover **83**.

(36) The lower cover **81** may be provided below the first cable slab **72**. For example, the lower cover **81** may have a thickness of about 0.75 mm. The lower cover **81** may have folding slits **85**. The folding slits **85** may expose a portion of the first cable slab **72**. Each of the folding slits **85** may have a rectangular shape when viewed in plan. Each of the folding slits **85** may have a vertical length of about 380 mm and a horizontal length of about 10 mm. The folding slits **85** may have therebetween a distance of about 50 mm to about 100 mm. In addition, the lower cover **81** may have a lower adhesive layer **88** on an edge of a top surface thereof. The lower adhesive layer **88** may include epoxy, but the present inventive concept is not limited thereto.

(37) The upper cover **83** may be provided on the first cable slab **72**. For example, the upper cover **83** may have a thickness of about 0.75 mm. The upper cover **83** may have folding slits **85**. The upper cover **83** may have an upper adhesive layer **89** on an edge of a bottom surface thereof. The upper adhesive layer **89** may include epoxy.

(38) The lower cover **81** and the upper cover **83** may be replaced in case of emergency or worn out. A knife (not shown) or a pair of scissors (not shown) may be used to separate worn-out lower and upper covers **81**, **83** may be separated from the first cable slab **72** or the second cable slab **74**. A pair of scissors may be provided on edges of the lower and upper covers **81** and **83** that are on an outer periphery of either the first cable slab **72** or the second cable slab **74**. After removing the worn-out lower and upper covers **81**, **83**, with new lower and upper covers **81**, **83** may be provided on the first cable slab **72** or the second cable slab **74**. A hand press tool may be utilized to compress and attach the lower cover **81** and the upper cover **83**. The hand press tool may heat and compress the lower cover **81** and the upper cover **83**.

(39) Each of the lower and upper covers **81**, **83** may have handlers **87**. The handlers **87** may be provided on edges of each of the lower and upper covers **81** and **83**. Each of the handlers **87** may have a tetragonal shape when viewed in plan. Each of the handlers **87** may have a vertical length of about 30 mm and a horizontal length of about 10 mm. The handlers **87** may be usefully employed to align the lower cover **81** and the upper cover **83**. In addition, the handlers **87** may be helpfully used to couple the lower cover **81** and the upper cover **83**.

(40) The handlers **87** may be removed when the lower and upper covers **81** and **83** are coupled, but the present inventive concept are not limited thereto. As shown In FIG. **5**, the handlers **87**, the lower adhesive layer **88**, and the upper adhesive layer **89** may be omitted on an edge of the first cable slab **72** between the lower cover **81** and the upper cover **83**.

(41) Although not shown in FIGS. **4** and **5**, the second cable cover **84** may include a lower cover **81**

and an upper cover **83**, but the present inventive concept is not limited thereto.

(42) FIG. **6** is a cross-sectional view, the reticle stage **60** depicted in FIG. **1**, according to an embodiment.

(43) Referring to FIG. **6**, the reticle stage **60** may further include a roller vehicle **90**. The roller vehicle **90** may be provided on the first and second cable slabs **72** and **74** that are curved. For example, the roller vehicle **90** may include a first roller **92**, a second roller **94**, and a connection bar **96**.

(44) The first roller **92** may be provided on the first cable slab **72** and the first cable cover **82**. For example, the first roller **92** may include a spring roll. Alternatively, the first roller **92** may include a bearing roller, but the present inventive concept is not limited thereto. The first roller **92** may be provided on a curved portion of the first cable slab **72**. For example, the first roller **92** may include a spring roll.

(45) The second roller **94** may be provided on the second cable slab **74** and the second cable cover **84**. The second roller **94** may be provided on a curved portion of the second cable slab **74**. The second roller **94** may be the same as the first roller **92**. For example, the second roller **94** may include a spring roll.

(46) The connection bar **96** may connect the first roller **92** to the second roller **94**. The connection bar **96** may define a distance between the first roller **92** and the second roller **94**. The connection bar **96** may move together with the first and second rollers **92** and **94**.

(47) Therefore, the roller vehicle **90** may maintain a constant distance between the first cable slab **72** and the second cable slab **74**.

(48) FIG. **7** is a cross-sectional view of the roller vehicle **90** depicted in FIG. **6**, according to an embodiment.

(49) Referring to FIG. **7**, the roller vehicle **90** may be provided on and below each of the first cable slab **72** and the second cable slab **74**. The first roller **92** of the roller vehicle **90** may be provided on and below the first cable slab **72**. For example, the first roller **92** may include a first upper roller **91** and a first lower roller **93**.

(50) The first upper roller **91** may be provided on the first cable slabs **72** and the first cable cover **82**. The first upper roller **91** may be provided in curved or bent portions of the first cable slab **72** and the first cable cover **82**. The first upper roller **91** may have a diameter greater than that of the first lower roller **93**.

(51) The first lower roller **93** may be provided below the first cable slab **72** and the first cable cover **82**. The first lower roller **93** may support curved portions of the first cable slab **72** and the first cable cover **82**. When the first cable slab **72** and the first cable cover **82** become bent, the first lower roller **93** may reduce loads of the first cable slab **72** and the first cable cover **82** against the first lower plate **62**. In addition, the first lower roller **93** may reduce a contact area between the first cable cover **82** and the first lower plate **62**, thereby decreasing abrasion of the first cable cover **82**. A rotational direction of the first lower roller **93** may be opposite to that of the first upper roller **91**.

(52) The second roller **94** of the roller vehicle **90** may be provided on and below each of the second cable slabs **74** and the second cover **84**. For example, the second roller **94** may include a second upper roller **95** and a second lower roller **97**.

(53) The second upper roller **95** may be provided on the second cable slab **74** and the second cable cover **84**. The second upper roller **95** may be provided in curved portions of the second cable slab **74** and the second cable cover **84**. The second upper roller **95** may have a diameter greater than that of the second lower roller **97**.

(54) The second lower roller **97** may be provided below the second cable slab **74** and the second cable cover **84**. The second lower roller **97** may support curved portions of the second cable slab **74** and the second cable cover **84**. When the second cable slab **74** and the second cable cover **84** become bent, the second lower roller **97** may reduce loads of the second cable slab **74** and the second cable cover **84** against the first lower plate **62**. In addition, the second lower roller **97** may

reduce a contact area between the second cable cover **84** and the first lower plate **62**, thereby decreasing abrasion of the second cable cover **84**. A rotational direction of the second lower roller **97** may be opposite to that of the second upper roller **95** and identical to that of the first lower roller **93**.

(55) The first upper roller **91**, the first lower roller **93**, the second upper roller **95**, and the second lower roller **97** may be connected through the connection bar **96** to each other in a tetragonal shape. For example, the connection bar **96** may include a first level connection bar **98**, a first horizontal connection bar **99**, a second level connection bar **111**, and a second horizontal connection bar **113**.

(56) The first level connection bar **98** may connect the first upper roller **91** to the first lower roller **93**. The first level connection bar **98** may define a distance between the first upper roller **91** to the first lower roller **93**. The first level connection bar **98** may be inclined to the first lower plate **62**. Alternatively, the first level connection bar **98** may be perpendicular to the first lower plate **62**, but the present inventive concept is not limited thereto.

(57) The first horizontal connection bar **99** may connect the first upper roller **91** to the second upper roller **95**. The first horizontal connection bar **99** may define a distance between the first upper roller **91** to the second upper roller **95**. The first horizontal connection bar **99** may maintain a constant distance between curved portions of the first and second cable slabs **72** and **74**.

(58) The second level connection bar **111** may connect the second upper roller **95** to the second lower roller **97**. The second level connection bar **111** may define a distance between the second upper roller **95** to the second lower roller **97**. The second level connection bar **111** may be inclined relative to the first lower plate **62**. Alternatively, the second level connection bar **111** may be perpendicular to the first lower plate **62**, but the present inventive concept is not limited thereto.

(59) The second horizontal connection bar **113** may connect the second lower roller **97** to the first lower roller **93**. The second horizontal connection bar **113** may define a distance between the first lower roller **93** to the second lower roller **97**. The second horizontal connection bar **113** may adjust a contact area between the first cable cover **82** and the first lower plate **62** and may also adjust a contact area between the second cable cover **84** and the first lower plate **62**.

(60) According to the present inventive concept, the semiconductor fabrication apparatus **100** may be configured such that the roller vehicle **90** is used to minimize abrasion of the first cable cover **82** and the second cable cover **84**.

(61) FIG. **8** is a view illustrating the reticle stage **60** depicted in FIG. **1**, according to an embodiment.

(62) Referring to FIG. **8**, the reticle stage **60** may further include first outer plates **61** and first inner plates **63**. The first actuator **66** may be configured identically to that of FIG. **2**.

(63) The first outer plates **61** may be provided outside opposite sidewalls of the first actuator **66**. The first outer plates **61** may be provided on outer peripheries of the first cable cover **82** and the second cable cover **84**.

(64) The first inner plates **63** may be provided between the first outer plates **61** and the opposite sidewalls of the first actuator **66**. The first inner plates **63** may be parallel to the first outer plates **61**.

(65) The first cable cover **82** and the second cable cover **84** may be provided between the first outer plates **61** and the first inner plates **63**. The first cable cover **82** and the second cable cover **84** may be folded in a direction parallel to the first actuator **66**. For example, the first cable cover **82** and the second cable cover **84** may be folded in a horizontal direction. Although not shown in FIG. **8**, the first cable slab (see **72** of FIG. **3**) and the second cable slab (see **74** of FIG. **3**) may be folded in a horizontal direction.

(66) Referring again to FIG. **1**, the substrate stage **20** may be configured identically to the reticle stage **60** of FIGS. **2**, **3**, and **6** to **8**, but the present inventive concept is not limited thereto.

(67) FIG. **9** illustrates the semiconductor fabrication apparatus **100** according to an embodiment of the present inventive concept.

(68) Referring to FIG. 9, a semiconductor fabrication apparatus **100** of the present inventive concept may be an extreme ultraviolet (EUV) exposure device. For example, the semiconductor fabrication apparatus **100**, according to the present embodiment, may further include a chamber **12**.

(69) The chamber **12** may envelope the light source **10**, the substrate stage **20**, the illumination optical system **30**, the projection optical system **40**, the reticle **50**, and the reticle stage **60**. The chamber **12** may store or hermetically seal plasma (not shown) of the light source **10**.

(70) The light source **10** may be provided on one side of a lower portion of the chamber **12**. The light **10** may generate either plasma or light **110** of plasma beam. For example, the light source **10** may produce the light **110** by providing pump light to liquid metal droplets of tin (Sn), xenon (Xe), titanium (Ti), or lithium (Li). The light **110** may include an extremely ultraviolet (EUV) beam having a wavelength of about 13.5 nm.

(71) The substrate stage **20** may be provided on a center of the lower portion of the chamber **12**. The substrate stage **20** may be provided below the illumination optical system **30** and the projection optical system **40**. The substrate stage **20** may receive and move the substrate W.

(72) FIG. **10** illustrates the substrate stage **20** of the semiconductor fabrication apparatus as depicted in FIG. **9**.

(73) Referring to FIG. **10**, the substrate stage **20** may include a second lower plate **22**, a second upper plate **24**, a second actuator **26**, a chucking plate **28**, a third cable slab **76**, and a third cable cover **86**.

(74) The second lower plate **22** may be provided below the second upper plate **24**, the second actuator **26**, and the chucking plate **28**. The second lower plate **22** may be connected to the controller **102**. Alternatively, the controller **102** may be separated from the second lower plate **22**, but the present inventive concept is not limited thereto.

(75) The second upper plate **24** may be provided on the second lower plate **22**. The second upper plate **24** may be parallel to the second lower plate **22**. The second upper plate **24** may be utilized as a cover plate.

(76) The second actuator **26** may be coupled to the second upper plate **24**. The second actuator **26** may move along a second guide rail (not shown). The second guide rail may be parallel to the second upper plate **24**. The second actuator **26** may support the chucking plate **28**. The second actuator **26** may drive the chucking plate **28** to move upwards, downwards, leftwards, and rightwards.

(77) Although not shown in FIG. **10**, the second actuator **26** may include a second long stroke actuator and a second short stroke actuator. The second long stroke actuator may be provided below the second short stroke actuator. The second long stroke actuator may be wider than the second short stroke actuator. The second long stroke actuator may drive the second short stroke actuator to move in a horizontal direction. The second short stroke actuator may be provided on a center of the second long stroke actuator. The second short stroke actuator may have a substrate chuck in a center thereof. The substrate W may be provided on the substrate chuck. The first short stroke actuator may drive the substrate W to move in a horizontal direction (e.g., X and Y directions).

(78) The chucking plate **28** may be provided on the second upper plate **24** and the second actuator **26**. The chucking plate **28** may receive the substrate W. The chucking plate **28** may fix the substrate W, and the second actuator **26** may drive the chucking plate **28** to move the substrate W.

(79) The third cable slab **76** may connect the second actuator **26** to the controller **102**. The third cable slab **76** may be provided between the second lower plate **22** and the second upper plate **24**. The third cable slab **76** may be curved or bent. A curved portion of the third cable slab **76** may move together with the second actuator **26**. Although not shown in FIG. **10**, the third cable slab **76** may be provided in plural in a horizontal direction (e.g., X and Y directions).

(80) The third cable cover **86** may surround an outer circumferential surface of the third cable slab **76**. The third cable cover **86** may protect the third cable slab **76**. The third cable cover **86** may be resistant to plasma. Although not shown, the third cable cover **86** may include a lower cover and an

upper cover. The lower cover may be provided below the third cable slab **76**. The upper cover may be provided on the third cable slab **76**. The lower and upper covers may be coupled and/or attached to each other on an outer periphery of the third cable slab **76**. Each of the lower and upper covers may have folding slits. The third cable cover **86** may be provided in plural in a horizontal direction (e.g., X and Y directions).

(81) Referring again to FIG. **9**, the illumination optical system **30** may be provided between the light source **10** and the reticle **50**. The illumination optical system **30** may reflect the light **110** on the reticle **50**. The illumination optical system **30** may include a plurality of flat mirrors.

(82) The projection optical system **40** may be provided between the reticle **50** and the substrate stage **20**. The projection optical system **40** may be configured such that the light **110** is reflected to project an image of the reticle **50** on the substrate **W**. The projection optical system **40** may include one or more of concave mirrors, convex mirrors, and flat mirrors, but the present inventive concept is not limited thereto.

(83) The reticle **50** may be provided in a central upper portion of the chamber **12**. Although not shown, the reticle **50** may be provided on the reticle stage **60**, while being received in a reticle stocker in the chamber **12**. The reticle **50** may reflect the light **110** on the projection optical system **40**.

(84) The reticle stage **60** may be provided in a central upper portion of the chamber **12**. The reticle stage **60** may fix the reticle **50**. The reticle stage **60** may be configured identically to the substrate stage **20**. In addition, the reticle stage **60** may be configured identically to that of FIGS. **2**, **3**, and **6** to **8**, but the present inventive concept is not limited thereto.

(85) According to the present inventive concept, the semiconductor fabrication apparatus **100** may be configured such that the first cable cover **82**, the second cable cover **84**, and the third cable cover **86** are used to prevent or minimize abrasion of the first cable slab **72**, the second cable slab **74**, and the third cable slab **76**.

(86) As discussed above, a semiconductor fabrication apparatus according to some embodiments of the present inventive concept may be configured such that a cable cover surrounding a cable slab is used to prevent or minimize abrasion of the cable slab.

(87) The above descriptions are specific examples for practicing the present inventive concept. The present inventive concept will include not only the embodiments described above but also embodiments that can be easily or simply changed in design. In addition, the present inventive concept will also include technique that can be easily modified and implemented using the embodiments described above.

Claims

1. A substrate fabrication apparatus, comprising: a light source configured to emit light; a substrate stage configured to receive a substrate exposed to the light; a reticle disposed between the substrate stage and the light source; and a reticle stage configured to receive the reticle, wherein the reticle stage includes: lower plate; an upper plate disposed on the lower plate; an actuator connected to the lower plate and configured to move in a direction parallel to the upper plate; a first cable slab disposed between the upper plate and the lower plate, the first cable slab connected to a first side of the actuator; and a first cable cover surrounding an outer periphery of the first cable slab, the cable cover configured to contact the lower plate when the first cable slab is bent.

2. The substrate fabrication apparatus of claim 1, wherein the first cable cover includes: a lower cover disposed below the first cable slab; and an upper cover disposed on the first cable slab and on an edge of the lower cover on an outer periphery of the first cable slab.

3. The substrate fabrication apparatus of claim 2, wherein each of the upper cover and the lower cover includes a plurality of folding slits that expose portions of the first cable slab.

4. The substrate fabrication apparatus of claim 3, wherein each of the plurality of folding slits has a

rectangular shape.

5. The substrate fabrication apparatus of claim 1, wherein the reticle stage further includes: a second cable slab disposed between the upper plate and the lower plate, the second cable slab connected to a second side of the actuator; and a second cable cover surrounding an outer periphery of the second cable slab, the second cable cover configured to contact the lower plate when the second cable slab is bent.

6. The substrate fabrication apparatus of claim 5, wherein the reticle stage further includes a roller vehicle disposed on the first cable slab and the second cable slab, the roller vehicle configured to move together with the actuator.

7. The substrate fabrication apparatus of claim 6, wherein the roller vehicle includes: a first roller disposed on the first cable cover; a second roller disposed on the second cable cover; and a connection bar configured to connect the first roller to the second roller.

8. The substrate fabrication apparatus of claim 7, wherein the first roller includes: a first upper roller disposed on the first cable cover; and a first lower roller disposed below the first cable cover.

9. The substrate fabrication apparatus of claim 8, wherein the second roller includes: a second upper roller disposed on the second cable cover; and a second lower roller disposed below the second cable cover.

10. The substrate fabrication apparatus of claim 9, wherein the connection bar includes: an upper connection bar disposed between the first upper roller and the second upper roller; a first connection bar disposed between the first upper roller and the first lower roller; a lower connection bar disposed between the first lower roller and the second lower roller; and a second connection bar disposed between the second upper roller and the second lower roller.

11. A semiconductor fabrication apparatus, comprising: a chamber; a light source in the chamber configured to emit light; a substrate stage configured to receive a substrate exposed to the light; a reticle disposed between the substrate stage and the light source; and a reticle stage configured to receive the reticle, wherein the substrate stage includes: a lower plate; an upper plate disposed on the lower plate; an actuator connected to the lower plate moving in a direction parallel to the upper plate; a cable slab between the upper plate and the lower plate, the cable slab being connected to a first side of the actuator; and a cable cover surrounding an outer periphery of the cable slab, the cable cover configured to contact the lower plate when the cable slab is bent.

12. The semiconductor fabrication apparatus of claim 11, wherein the cable cover includes Teflon.

13. The apparatus of claim 11, wherein the cable cover includes: a lower cover disposed below the cable slab; and an upper cover disposed on the cable slab and an edge of the lower cover on an outer periphery of the cable slab.

14. The substrate fabrication apparatus of claim 13, wherein each of the upper cover and the lower cover has a plurality of folding slits exposing portions of the cable slab.

15. The substrate fabrication apparatus of claim 14, wherein each of the folding slits has a rectangular shape.

16. A semiconductor fabrication apparatus, comprising: a light source configured to emit light; a substrate stage configured to receive a substrate exposed to the light emitted from the light source; a reticle disposed between the substrate stage and the light source; and a reticle stage configured to receive the reticle, wherein the reticle stage includes: a first lower plate; a first upper plate disposed on the first lower plate; a first actuator connected to the first lower plate and configured to move in a direction parallel to the first upper plate; a first cable slab disposed between the first upper plate and the first lower plate, the first cable slab connected to a first side of the first actuator; a first cable cover surrounding an outer periphery of the first cable slab configured to contact the first lower plate when the first cable slab is bent; a second cable slab disposed between the first upper plate and the first lower plate connected to a second side of the first actuator; a second cable cover surrounding an outer periphery of the second cable slab configured to contact the first lower plate when the second cable slab is bent; and a roller vehicle disposed on the first cable slab and the

second cable slab, the roller vehicle configured to move together with the first actuator.

17. The semiconductor fabrication apparatus of claim 16, wherein the roller vehicle includes: a first roller disposed on the first cable cover; a second roller disposed on the second cable cover; and a connection bar configured to connect the first roller to the second roller.

18. The semiconductor fabrication apparatus of claim 17, wherein the first roller includes: a first upper roller disposed on the first cable cover; and a first lower roller disposed below the first cable cover.

19. The semiconductor fabrication apparatus of claim 18, wherein the second roller includes: a second upper roller disposed on the second cable cover; and a second lower roller disposed below the second cable cover.

20. The semiconductor fabrication apparatus of claim 19, wherein the connection bar includes: an upper connection bar disposed between the first upper roller and the second upper roller; a first connection bar disposed between the first upper roller and the first lower roller; a lower connection bar disposed between the first lower roller and the second lower roller; and a second connection bar disposed between the second upper roller and the second lower roller.
